Kenya Math Camp Report, May 2023

Location: Kongoni Community Library, Kakamega County, Kenya.

Dates: 1th May - 5th May 2023



Introduction

The Kenya Math Camp 2023, held at Kongoni Community Library in Likuyani Constituency, Kakamega County in Kenya, was a transformative event that aimed to provide students with an immersive and interactive learning experience in mathematics. With the participation of 50 students from 11 secondary schools, the camp fostered an inclusive environment where students could engage in active learning, critical thinking, and practical application of mathematical concepts. The diverse group of attendees, comprising 54% males and 46% females aged between 15 and 22, brought together students from different academic levels, including form fours, form threes, form twos, and form ones. While 13 students had prior experience attending a math camp, the majority of participants, 74%, had the opportunity to attend for the first time.

Among the 10 facilitators present at this year's math camp, it is noteworthy that two of them were former math camp alumni who actively participated in the planning and execution of the camp. Their valuable contribution and experience as former participants added a unique

perspective to the camp. Special thanks are extended to *Elphas Ogong'o, Samuel Okoth, Owen Mulinya Kizito, Talu Lucian Mayabi, Geoffrey Mboya, Feleria Adinda, Emily Flemming, Jeff Goodman, Debora Nyaboke, Faith Mbugua, Jediddah Ndusya, Michael Chariga, Titus Kinuthia, Cheryl Muvoka,* and *Nelson Kassim* for their dedication and support as facilitators. Their efforts greatly contributed to the success and positive experience of the math camp.

Throughout the camp, students acquired valuable skills that will undoubtedly benefit their academic pursuits and future endeavors. The engaging activities, interactive sessions, and collaborative learning opportunities provided students with a solid foundation in various mathematical topics. By promoting critical thinking, problem-solving, and teamwork, the camp empowered students to approach mathematics with confidence and enthusiasm. The success of the Kenya Math Camp 2023 serves as a testament to the importance of creating dynamic learning environments that inspire and nurture students' mathematical abilities.

Themes

The camp focused on various themes, including Programming and Robotics, Modeling, Mathematical Thinking, Chess and Mathematics, Data Science, Physical Activities, and Games and Puzzles. The main theme of the camp was “Exploring the Beauty of Mathematics for the 22nd Century”. All themes integrated an effective use of technology, handson activities, effective questioning, inquiry-based instruction, cooperative learning and encouraging active participation of students.

Programming and Robotics enabled students to develop logical and linguistic skills through active learning methodologies such as project-based learning and problem-solving. Sessions on *modeling* encouraged students to address real-world issues using mathematical models. *Data Science* covered various aspects of data analysis, including data storytelling, data cleaning, data representation, and data visualization.

Mathematical Thinking sessions focused on developing critical thinking skills through engaging activities like chess, city skyline, and the Tower of Hanoi puzzle. Students learned problem-solving strategies and the systematic approach to tackling mathematical challenges. *Physical Activities* incorporated fun games that reinforced mathematical concepts such as counting, patterns, and computing techniques. *Games and Puzzles* provided students with opportunities to engage in logic puzzles, brain teasers, and number challenges. They were encouraged to solve puzzles of varying difficulty levels, fostering problem-solving skills and critical thinking. Students also had the chance to take on the role of facilitators, supporting others in puzzle-solving.

Programming and Robotics

The programming and robotics sessions at the maths camp aimed to develop students' logical and linguistic skills while providing a real-world setting to apply curricular concepts. This theme embraced active learning methodologies such as project-based learning, problem-solving, gamification, and cooperative learning, enabling students to take charge of their education and engage in robotics within an active learning environment.

Human Robot Game and Blockly Game: The hands-on sessions began with the Human Robot Game, where students explored how computers work and "think." They experimented with writing programs in a format understandable by a human robot, gaining insights into creating effective computer programs for specific robots. This activity fostered computational thinking among the students, encouraging them to plan, problem-solve, and analyze information in a logical manner, similar to how a computer processes data. The process involved breaking down problems into smaller components, identifying patterns, and formulating step-by-step solutions.

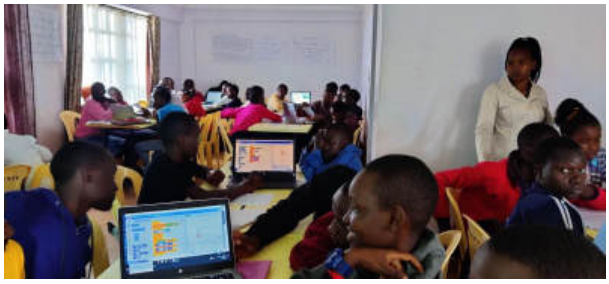
Block-Based Programming with Blockly: The sessions introduced students to block-based programming, a programming language where instructions are issued by dragging and dropping blocks. Specifically, Blockly, a block-based programming language developed by Google, was utilized. It employed graphical blocks to represent code concepts such as variables, loops, logical expressions, and more. Blockly allowed students to focus on learning programming fundamentals without the need to worry about syntax.

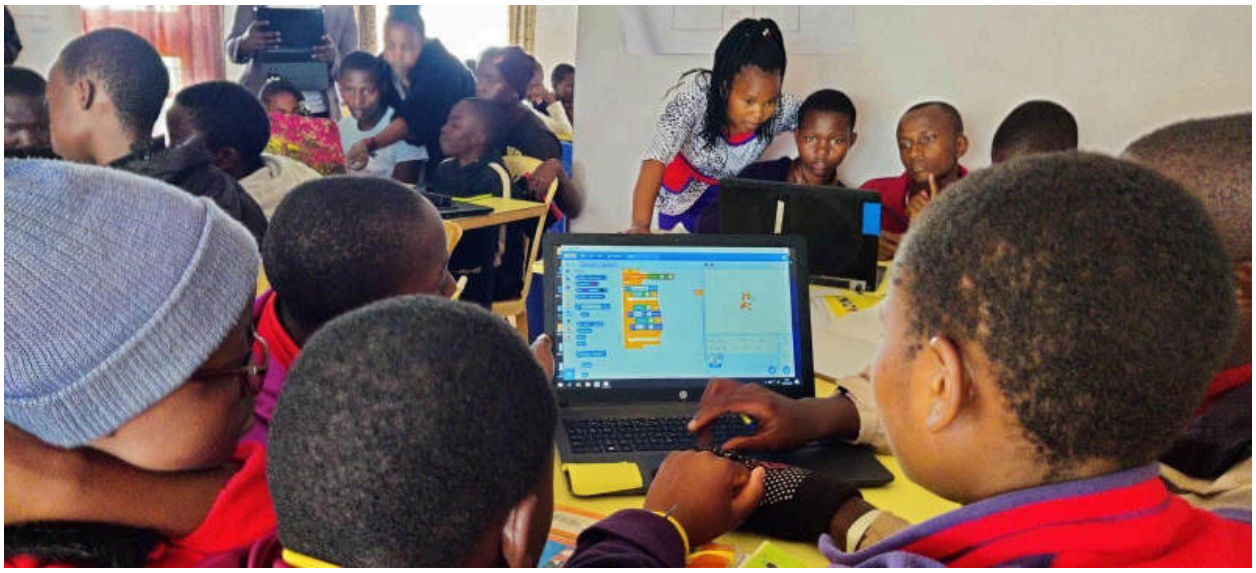
Edison Robot: The sessions also included the use of Edison robots. Edison robots are programmable robots designed to introduce students to robotics and coding. Students had the opportunity to interact with these robots and learn how to program them to perform various tasks and navigate through different challenges. Edison robots provided a hands-on experience for students to apply their programming skills and see the direct impact of their instructions on the robot's behavior. This aspect of the sessions further enhanced students' understanding of robotics and allowed for a more comprehensive exploration of programming concepts.

Creative Coding with Scratch: As the sessions progressed, there was an increased focus on creative coding using Scratch. Students were encouraged to code, learn, and explore their own ideas. They developed flowcharts for games and then implemented them using Scratch, a visual programming language. The students had the opportunity to add more features and personalize their games, fostering creativity and critical thinking.

The programming and robotics sessions at the Maths camp provided students with a valuable opportunity to enhance their logical and linguistic skills while applying curricular concepts in a real-world context. The incorporation of hands-on activities enabled students to develop computational thinking, problem-solving abilities, and programming fundamentals. By making

learning interactive and enjoyable, the sessions successfully engaged students in the world of programming and robotics, empowering them to explore their own ideas and develop valuable skills for the future.





Fi: *Showing students coding to learn in scratch and using Edison Robot*

Modeling

The sessions on modeling served as both an introduction to higher learning math and an inspiration for how to address problems using mathematical models. It's fun to introduce students to real-world problems, solved through mathematical models incorporating the use of technology to apply logical reasoning, forecast, and develop control measures. Giving students the chance to recognize some societal issues as well as attempt to develop mathematical models based on evidence. The sessions on modeling included:

Math for Climate Action: Recent discussions have focused heavily on climate change, so having a session on it at the camp was beneficial. Students expect to learn problem-solving and critical thinking skills, therefore using secondary data on carbon dioxide emissions since 1900, they were able to develop a model that might forecast the level of carbon dioxide in the near future. Geogebra served as our modeling tool. We examined the value of models and concentrated on two in particular. Since models are meant to predict, our model allowed us to make predictions.

Similar models aid in the development of control measures. Planting trees was one of the controls that stood out. As actions for the session, we were able to plant three trees at the conclusion of the camp. Students were also able to recognize societal issues through the camps and offer solutions based on mathematical models.



Mathematical Thinking

Through a series of eight activities—counting squares, frog's party, knock down puzzle, graph coloring, card game, twin prime, city skyline, and tower of Hanoi—the math camp emphasized the development of mathematical thinking skills. The campers were urged to use their critical thinking abilities to spot trends and create problem-solving plans. For the purpose of stimulating the campers' thinking and enhancing their mathematics skills, a new issue or subject was taught in each session. The campers left with a solid understanding of mathematics, organizational skills, and a methodical way of thinking that they could use to tackle a variety of issues.

Chess: Chess, a game with a rich history and intricate gameplay, has captivated thinkers and mathematicians alike for centuries. While it may be seen as a recreational activity, chess possesses several qualities that make it a valuable tool in the realm of mathematics. This report aims to explore the connection between chess and mathematics, as seen in the sessions run at the camp, highlighting the usefulness of chess as a mental exercise for mathematical reasoning, problem-solving, and strategic thinking.

The participants had the opportunity to be introduced to chess as a useful mental exercise, to help develop skills that are directly applicable to mathematical problem-solving and analytical thinking. Embracing the lessons and principles derived from chess can enhance one's mathematical abilities and contribute to a deeper understanding of various mathematical concepts.

The students were introduced to tic-tac-toe and connect-4 games, and reflected on games and strategies before being introduced to the game of Chess, as to cultivate cultures of individuals capable of critical thought and mathematical problem-solving. One of the main goals of the workshop was to learn the rules and put them to use. Despite having fewer chess boards, the students' curiosity was clear. We hope that the attendees will be able to participate in chess activities going forward and grow interested in them. Chess, despite being a game, possesses remarkable usefulness in the realm of mathematics. It engages logical reasoning, pattern recognition, strategic thinking, decision making, and aligns with game theory concepts.



City skyline: The City Skyline is a mathematical card game that involves arranging a set of cards in a specific pattern. The game begins with a shuffled set of numbered cards, and the player creates multiple piles of cards. In each turn, they take one card from each pile to form a new pile. The process continues until a previously seen configuration is reached. The objective is to achieve a desired pattern, typically a consecutive sequence of numbers starting from 1. The game promotes critical thinking, problem-solving, and enjoyment.

Twin Primes: This session introduced learners to another set or type of number(s). It was also a great opportunity as learners explored the R-Instat cool calculator for the first time. Learners could identify the twin primes from the listed prime numbers but this was a bit tiresome, so we had to automatically generate the twin primes by use of the R-Instat cool calculator. The session can be found here.

Card Game- MOD 10 and 8: This session was a fun session where learners played with cards using the given set of rules. This is a card game where you do basic mental calculations and race your opponent (2-4 players). Using Ace -10 cards only. Additionally, learners were being

introduced to the modulus concept as the majority of them were interacting with the "MOD" term for the first time. You can explore the rules of the game and more here.



Frogs Party: In this session, the facilitator used a modeling approach to introduce a problem to the class. They demonstrated the problem using three students as lily pads and sticky notes as frogs. The facilitator explained the game rules and emphasized that they would not provide a complete solution, leaving it as a task for the students.

The game involved placing frogs on lily pads and the objective was to transport all the frogs to a single lily pad where a party would be held. There were specific constraints for moving the frogs, including moving one frog at a time, placing a frog on top of another frog, and making steps according to the number of frogs present on a lily pad.

By using this hands-on approach, the facilitator engaged the students in the problem-solving process and encouraged them to think critically about the game.



The Tower of Hanoi: The Tower of Hanoi is a classic mathematical puzzle involving the movement of a stack of disks between three pegs. The puzzle adheres to three simple rules: one disk can be moved at a time, disks are placed on top of other disks, and larger disks cannot be placed on top of smaller disks.

At the Math Camp, students tackled the Tower of Hanoi puzzle using two methods: the inductive method and the iterative method. The inductive method began with the base case of moving a single disk, and then students used recursion to solve the puzzle for increasing numbers of disks. They broke down the problem into smaller sub-problems, moving disks between pegs with the help of an auxiliary peg.

Alternatively, the iterative method involves a loop that iterates through each disk. Students identified patterns and sequences of moves for each number of disks, adjusting the source, destination, and auxiliary pegs accordingly. They repeated the loop until all disks were successfully transferred to the destination peg.

Both methods allowed students to solve the Tower of Hanoi puzzle by recognizing patterns, leveraging previous solutions, and systematically moving the disks. By applying these methods, students gained a deeper understanding of problem-solving, patterns, and recursion.

Data Science

The math camp included several data science sessions that covered various topics such as data storytelling, data cleaning, data survey, and data representation and visualization. These sessions

aimed to familiarize participants with the entire data process, from gathering and cleaning data to visualizing and analyzing the results. Participants also learned how to use data effectively to support their arguments and make informed decisions.

Introduction to data stories: During the lesson, the main focus was on the initial step of data stories, which involves gathering ideas, information, and developing survey questions. To facilitate this, we utilized the Kenya 2019 census data as an example to depict the life of an average Kenyan. Throughout the session, we emphasized the three crucial components of effective survey questions: the question stem, response choices, and instructions.

By the end of the session, the students actively collaborated and generated ideas and information they wanted to learn from one another. Together, they formulated survey questions to collect relevant data. These survey questions were compiled and uploaded to a Google Form, which served as the platform for data collection and analysis. Notably, students from Kenya and London participated in filling out the questionnaire form, fostering cultural exchange and global learning. Subsequently, data cleaning procedures were employed using the acquired data from the session.

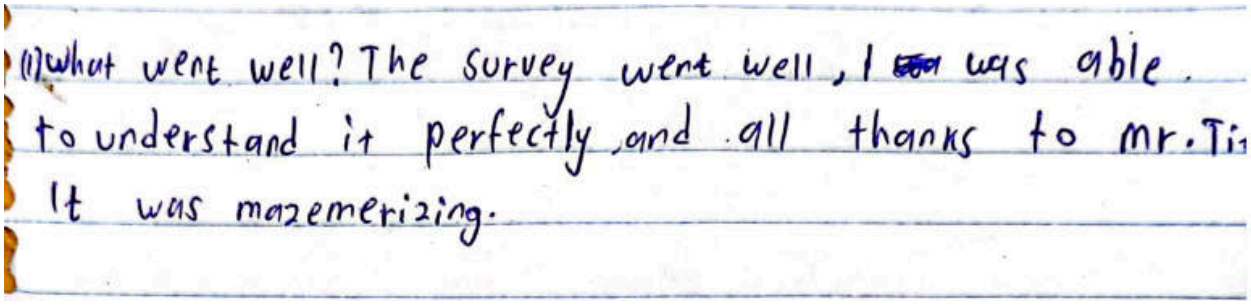


Fig: Sample journal on a student's reflection on the session.

Data Cleaning: This session was done where the data collected during maths camp entry survey session was fixed by removing incorrect, corrupted, incorrectly formatted, duplicate or incomplete data within that dataset. This was done by students with the help of instructions from facilitators. Data collected in a data entry survey was used to display how data cleaning is done.

Data Representation: In this session, students utilized a cleaned dataset of climatic/weather data for Kakamega County in April 2023. After completing data collection and data cleaning sessions, they proceeded to analyze the data. The students employed various visualization techniques such as 2-D pie charts, 3-D pie charts, and bar graphs to represent the data effectively. By applying these visualization methods, the students gained insights and a better understanding of the climatic patterns in Kakamega County during the specified time period.

Data Visualisation: This session came after data collection, data cleaning and data representation. CODAP was our data visualization tool.

Physical Activities

The camp had programs that involved fun games that everyone was encouraged to participate in. These games were played both indoors and outdoors and involved physical activities, which helped the campers learn different mathematical concepts such as counting, patterns, and computing techniques. The physical activities included a variety of games

Concentration game: Participants in this session engaged in a fun game aimed at improving focus, coordination, and memory. They formed a circle and followed rhythmic actions of tapping legs and clapping. The game also involved verbal cues where the leader said their own name and another person's name, requiring participants to remember and repeat the pattern accurately. The activity provided cognitive benefits and fostered collaboration among the group.



Fig: *Showing students playing concentration game*

Find the ball: Each team must direct a member of their team who is wearing a blindfold towards a particular object in this group activity. The activity promotes teamwork that works well, as well as communication and individual concentration.

Card game: Each team is assigned a hand of cards, ranging from Ace to Ten. The 10 cards for each team are dealt face down onto a table in front of them at random. One by one, the players come up and flip a card. If it's not the Ace, they flip it over one more time and dash back to their group, where the next player takes a turn. Once it has been revealed, they can lay the ace face up at the head of the table. Card two is then required, and so forth. Play continues until one side has correctly turned all of its cards face up. Here is one of the student's reflection on the activity:

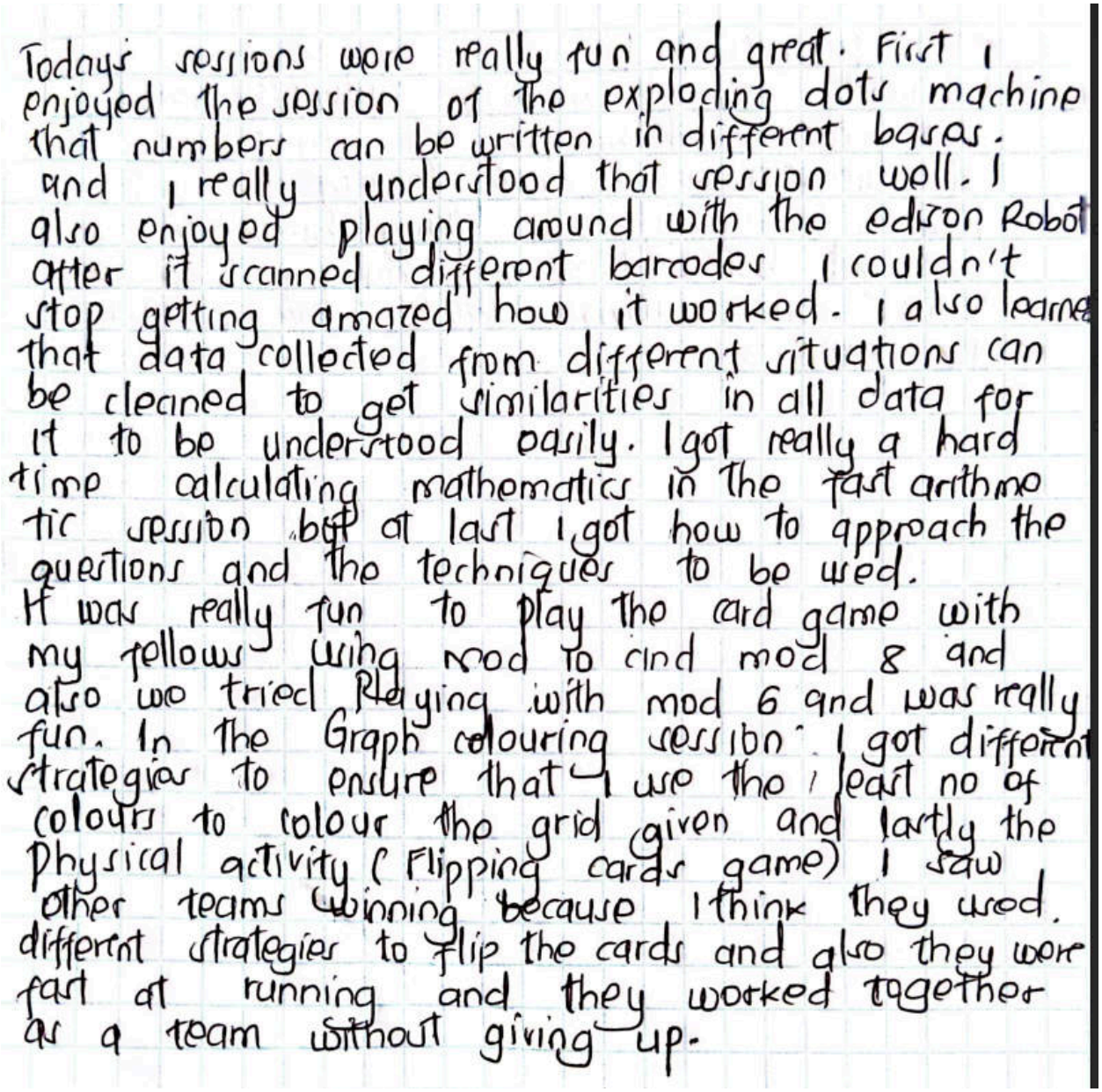

Todays sessions were really fun and great. First I enjoyed the session of the exploding dots machine that numbers can be written in different bases. and I really understood that session well. I also enjoyed playing around with the edison Robot after it scanned different barcodes I couldn't stop getting amazed how it worked. I also learned that data collected from different situations can be cleaned to get similarities in all data for it to be understood easily. I got really a hard time calculating mathematics in the fast arithmetic session but at last I got how to approach the questions and the techniques to be used. It was really fun to play the card game with my fellows using mod to find mod 8 and also we tried Playing with mod 6 and was really fun. In the Graph colouring session I got different strategies to ensure that I use the least no of colours to colour the grid given and lastly the Physical activity (Flipping cards game) I saw other teams winning because I think they used different strategies to flip the cards and also they were fast at running and they worked together as a team without giving up.

Fig: Showing students journal entry on the games

Listen and React: Three cones of the same color are given to each squad, three on each side, to the right and left. One of the team members is chosen to represent them. Since it's a two-player game, they both walk closer to the cones. They are expected to pay attention, and when a color is mentioned, they are expected to react by choosing the corresponding cone. whoever picks first wins.



Fig: *Showing maths camp participants participating in the game*

Games and Puzzles

The math camp provided students with a range of engaging activities and puzzles. These activities covered various mathematical topics, such as logic puzzles, time distance math, brain teasers, probability concepts, and number challenges. Throughout the camp, students were encouraged to solve puzzles of different difficulty levels that were displayed on the walls. They could work on these puzzles during their free time and even take them home as assignments. The students not only solved puzzles themselves but also acted as facilitators, assisting others in solving puzzles. This role allowed them to develop problem-solving skills, foster collaboration, and deepen their understanding of mathematical concepts. The math camp provided a valuable opportunity for students to engage in critical thinking, support each other's learning, and enhance their overall mathematical abilities.

Math competition: The inclusion of a math competition for our younger campers this year provided them with an opportunity to showcase their puzzle-solving skills within their respective houses. The math competition served multiple purposes, benefiting participants in various ways.

To begin with, the competition presented a challenge that engaged and stimulated the students' mathematical abilities. By offering intriguing puzzles and problems, the competition encouraged participants to develop their problem-solving skills and enhance their mathematical reasoning. Through the process of analyzing and solving these math challenges, students were prompted to think critically and explore different approaches to arrive at the correct solutions.

Moreover, the math competition played a crucial role in promoting a love for mathematics among the participants. By highlighting the beauty and practical applications of math beyond the

classroom, students gained a deeper appreciation for the subject. The competition showcased how math extends beyond formulas and equations, demonstrating its dynamic nature and relevance in real-life situations. This exposure to the broader aspects of mathematics instilled a sense of curiosity and enthusiasm, motivating students to further explore and excel in the subject.

Furthermore, the math competition fostered healthy competition and teamwork among the participants. As students competed against each other, they were motivated to push their boundaries and strive for excellence. The friendly rivalry encouraged them to work harder, improve their skills, and achieve higher results. Additionally, the competition emphasized collaboration as participants engaged in discussions, shared strategies, and learned from one another. These collaborative efforts not only enhanced the overall learning experience but also developed essential teamwork skills that are valuable for their future academic and professional pursuits.

The graph below displays the distribution of performance across the different houses, providing a visual representation of the competition. The graph showcases the varying levels of achievement among the participants, demonstrating the range of skills and capabilities showcased during the competition.

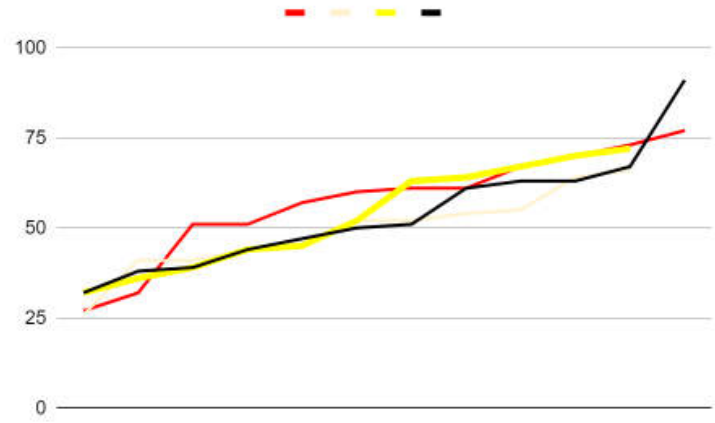


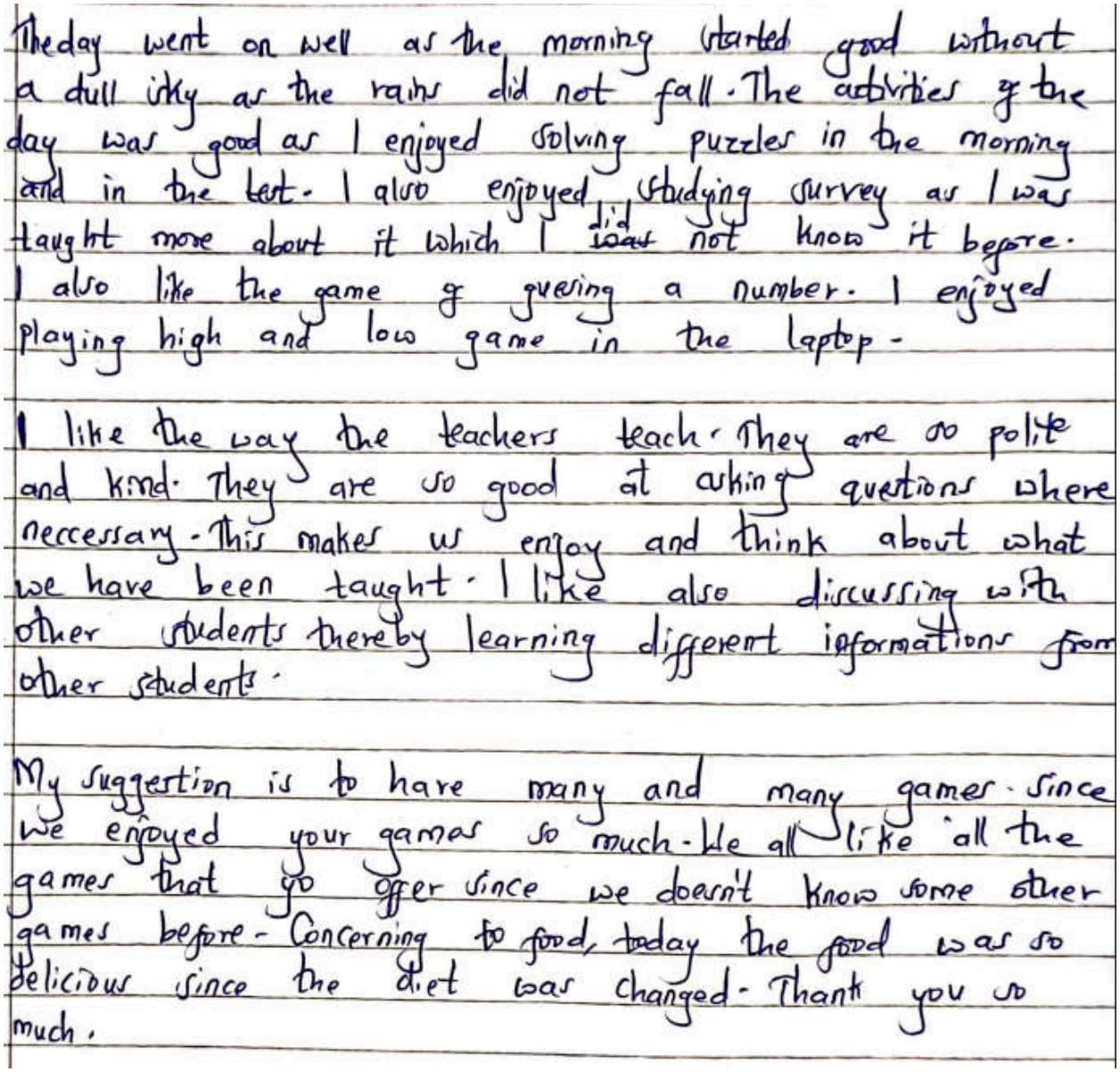
Fig: *Graph showing distribution of performance in different houses*

Quick Math: This session involved Eight mathematics problems given to participants every day for ten minutes before lunch. Renamed “Quick Math” During this session, the students went into their houses, and each house received points for each accurate response. The lesson challenged students to utilize their minds alone, no pens,no calculators. The questions displayed in this session normally build on each other concept-wise from day one.

Journals

The students at the math camp were encouraged to keep journals and reflect on their experiences. The journals provided valuable feedback for the organizers, covering what went well, what they didn't understand, and suggestions for improvement. The journals from each day of the camp showed students' enjoyment of certain activities and challenges with specific concepts. They suggested more time and examples for better understanding. The reflections also highlighted the benefits of group work and collaboration. Students acknowledged their progress and expressed appreciation for facilitators' support. The journals demonstrated the students' learning and provided insights for enhancing future camps. Overall, the journals were a valuable tool for improving the math camp experience for everyone involved.

Here is a sample of one of the student’s entry in journal:


The day went on well as the morning started good without a dull icky as the rains did not fall. The activities of the day was good as I enjoyed solving puzzles in the morning and in the last. I also enjoyed studying survey as I was taught more about it which I did not know it before. I also like the game of guesing a number. I enjoyed playing high and low game in the laptop.

I like the way the teachers teach. They are so polite and kind. They are so good at asking questions where neccessary. This makes us enjoy and think about what we have been taught. I like also discussing with other students thereby learning different informations from other students.

My suggestion is to have many and many games. Since we enjoyed your games so much. We all like all the games that yo offer since we doesn't know some other games before. Concerning to food, today the food was so delicious since the diet was changed. Thank you so much.

Fig: Sample Journal entry

Houses

The math camp consisted of four houses that encouraged collaboration and inter-house competitions. The houses were named after notable mathematicians and had distinctive colors: George Musengi Saitoti (red), Pythagoras of Samos (cream), Euphemia Haynes (yellow), and Verdiana Grace Masanja (black).

George Musengi Saitoti, a Kenyan mathematician and economist, conducted research in topology.

Pythagoras of Samos, known for the Pythagorean theorem, was not just a mathematician but also a spiritual leader. His followers believed he was sent from Heaven, and math was considered a religious experience for them.

Euphemia Haynes, an American mathematician and educator, holds the distinction of being the first African American woman to earn a PhD in mathematics.

Verdiana Grace Masanja, from Tanzania, became the first woman to obtain a Ph.D. in mathematics in her country. She achieved this milestone at the age of 33 in 1986 at the Technical University of Berlin.

Next Steps

Our next steps involve recruiting international volunteers, preparing the new camp location, enhancing the familiar location, collaborating with Bonn volunteers, establishing year-round support, forging collaborations and partnerships, conducting research and evaluation, scaling up our efforts, and providing more volunteer and teacher training as Maths game ambassadors. By taking these actions, we aim to create a sustainable and impactful Maths Camp that nurtures mathematical literacy, inspires a love for learning, and empowers students to excel in mathematics. As a next step for the Maths Camp, we will focus on the following key actions:

Volunteer Recruitment:
We have actively invited interest from international volunteers and aim to recruit a minimum of six participants.

New Location Preparation:
For the upcoming August camps, we have decided to run two separate camps, with one in a new location. To ensure a smooth transition, we will conduct site visits to assess the facilities available. Collaborating with local partners, we will ensure that the venue meets the necessary requirements, including suitable accommodation, classrooms, and recreational spaces. This will provide an engaging and conducive environment for the students' learning experience.

Previous Location Enhancement:
For the camp held in the familiar location, we recognize the importance of building upon previous successes and improving based on participant feedback. We will work closely with our team to enhance the learning environment by introducing new activities and incorporating

innovative teaching methods. Leveraging the experience gained from previous camps, we aim to deliver an even better and more impactful experience for the students.

Collaboration with Bonn Volunteers:
We are excited to engage in conversations with volunteers from Bonn and explore opportunities for collaboration. This exchange of knowledge and expertise will foster mutual learning and growth in the field of education. We will actively seek input from Bonn volunteers and explore ways to integrate their projects and initiatives, such as the support for Chess, into the Maths Camp. This collaboration will broaden the learning opportunities available to the students and enhance their overall experience.

Year-round Maths Club Support:
To extend the impact of the Maths Camp, we understand the importance of ongoing engagement to support initiating maths clubs. We are working to establish a year-round support system, providing online resources, mentorship programs, and virtual learning opportunities for students who have attended the camp. This sustained support will reinforce their mathematical skills, foster their love for learning, and ensure continuous growth even beyond the duration of the camp.

Collaboration and Partnerships:
We recognize the power of collaboration and partnerships in advancing math education. We will actively seek collaborations with educational institutions, government bodies, and corporate organizations. By pooling our resources and expertise, we can create sustainable initiatives that reach a larger audience. These collaborations will enable us to expand the reach and impact of the Maths Camp, making a difference in the lives of more students across Kenya.

Research and Evaluation:
To ensure the effectiveness and continuous improvement of our programs, we are committed to conducting research and evaluation studies. By gathering valuable insights and measuring the impact of the Maths Camp, we can make data-driven decisions to enhance the learning experience for future participants. This will involve collecting feedback from students, volunteers, and educators, and analyzing the data to inform program enhancements and ensure optimal outcomes.

Scaling Up:
Building on the success of this year's camp, we have set our sights on reaching even more students in the coming years. By forging partnerships with additional schools and communities, we aim to extend the benefits of the Maths Camp to a larger number of students across Kenya. Through strategic planning and resource allocation, we will scale up our operations while maintaining the quality and impact of the camp.

Teacher Training:
Recognizing the crucial role of educators in shaping students' mathematical abilities, we plan to organize training programs for teachers. These workshops will equip them with effective teaching strategies, innovative approaches, and resources to enhance math instruction in their classrooms. By empowering teachers with the necessary skills and knowledge, we aim to create a multiplier effect, positively impacting a larger number of students beyond the Maths Camp itself.

Conclusion

The camp's focus on active learning, critical thinking, and practical application of mathematical concepts proved to be highly effective. Students of various academic levels, including 16 form fours, 18 form threes, 10 form twos, and 6 form ones, benefited from the engaging sessions and activities. It is worth noting that 13 students had prior experience attending a math camp, while the remaining 74% attended for the first time.

The success of the Kenya Math Camp 2023 can be attributed to the dedication of the organizers, the support of the Kongoni Community Library, and the enthusiasm and active participation of the students. The camp served as a testament to the transformative power of immersive and interactive learning experiences, solidifying its position as a catalyst for academic growth and personal development.

We hope this report has inspired you to develop your own teaching initiatives and learning activities. For more information and access to additional resources, please visit virtualmathscamp.com. We extend our sincere appreciation to the kind sponsors who made the Kenya Math Camp 2023 possible: Supporting African Maths Initiative (SAMI), University of Geneva, African Maths Initiative (AMI), INNODEMS, and Kongoni Community Library.